Exercises

# Logic Programming

https://moodle.aau.at/course/view.php?id=47556

University of Klagenfurt — Winter semester 2026

Exercise 1 (Demonstration: 21 October 2026)

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## Quiz

| | | Answer | |
|---|---|---|---|
| **1.** | Each positive logic program has at most one model. | **true** | **false** |
| **2.** | Each positive logic program has at most one answer set. | **true** | **false** |
| **3.** | Each normal logic program has at least one model. | **true** | **false** |
| **4.** | Each normal logic program has at least one answer set. | **true** | **false** |

## Exercise 1.1

Determine the answer set of the following positive logic programs:

1.1–a

```
a.
b :- e.
b :- f.
c :- b, f.
c :- d, g.
d.
e :- b, d.
g :- a, b.
g :- a, d.
```

1.1–b

```
likes(kate,fish).
likes(kate,soup).
likes(mike,beef).
likes(mike,cake).
likes(paul,duck).
likes(paul,pork).
likes(tina,beef).
likes(tina,soup).

% order(Z) :- likes(X,Z), likes(Y,Z), X != Y.
order(beef) :- likes(mike,beef), likes(tina,beef).
order(soup) :- likes(kate,soup), likes(tina,soup).

% happy(X) :- likes(X,Y), order(Y).
happy(kate) :- likes(kate,fish), order(fish).
happy(kate) :- likes(kate,soup), order(soup).
happy(mike) :- likes(mike,beef), order(beef).
happy(mike) :- likes(mike,cake), order(cake).
happy(paul) :- likes(paul,duck), order(duck).
happy(paul) :- likes(paul,pork), order(pork).
happy(tina) :- likes(tina,beef), order(beef).
happy(tina) :- likes(tina,soup), order(soup).
```

## Exercise 1.2

Consider the following sets of (true) atoms:

- $X_1 = \{\texttt{a}, \texttt{b}, \texttt{d}, \texttt{e}\}$
- $X_2 = \{\texttt{b}, \texttt{e}\}$

Decide which of the above sets of atoms are answer sets for the following normal logic programs:

1.2–a

```
a :- b.
b :- not c.
c :- not b.
d :- a, e.
e.
```

1.2–b

```
a :- not b.
a :- d.
b :- not c.
c :- a, not b.
d :- a.
d :- not c, not e.
e :- b, not a.
e :- d.
```

## Exercise 1.3

Reconsider the two logic programs in the previous exercise and find all their answer sets. (HINT: Avoid considering all of the possible interpretations.)

## Exercise 1.4

Determine the answer sets of the following normal logic programs:

1.4–a

```
canary(tweety).
penguin(tux).

bird(tweety) :- canary(tweety).
bird(tux) :- penguin(tux).

injured(tweety) :- not ok(tweety), bird(tweety).
ok(tweety) :- not injured(tweety), bird(tweety).
injured(tux) :- not ok(tux), bird(tux).
ok(tux) :- not injured(tux), bird(tux).

flies(tweety) :- ok(tweety), not penguin(tweety).
flies(tux) :- ok(tux), not penguin(tux).
```

1.4–b

```
num(1). num(2). num(3).

lucky(1) :- num(1), not unlucky(1).
unlucky(1) :- num(1), not lucky(1).
lucky(2) :- num(2), not unlucky(2).
unlucky(2) :- num(2), not lucky(2).
lucky(3) :- num(3), not unlucky(3).
unlucky(3) :- num(3), not lucky(3).

p :- unlucky(1), not p.
p :- unlucky(2), not p.
p :- unlucky(3), not p.
```

Exercises

# Logic Programming

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University of Klagenfurt — Winter semester 2026

Homework 1 (Correction: 28 October 2026)

## Exercise 1.1

Determine the answer set of the following positive logic programs:

1.1–a

```
a :- b, d.
a :- c, e.
b :- d, m.
b :- e, f.
c :- d, e.
d.
e :- d.
f :- b, c.
g :- d, f.
g :- e, h.
h :- c, k.
k :- a, e.
```

1.1–b

```
conn(amsterdam,brussels).
conn(brussels,cologne).
conn(cologne,amsterdam).

% tour(X,Z) :- conn(X,Y), conn(Y,Z).
tour(amsterdam,cologne)  :- conn(amsterdam,brussels),
                            conn(brussels,cologne).
tour(brussels,amsterdam) :- conn(brussels,cologne),
                            conn(cologne,amsterdam).
tour(cologne,brussels)   :- conn(cologne,amsterdam),
                            conn(amsterdam,brussels).

% finish(Y) :- tour(X,Y), conn(X,amsterdam).
finish(cologne)   :- tour(amsterdam,cologne),
                     conn(amsterdam,amsterdam).
finish(amsterdam) :- tour(brussels,amsterdam),
                     conn(brussels,amsterdam).
finish(brussels)  :- tour(cologne,brussels),
                     conn(cologne,amsterdam).
```

## Exercise 1.2

Determine the reducts for the sets of atoms given in comments, and decide which of the sets of atoms are answer sets for the following normal logic programs:

1.2–a

```
%*
X1 = {}
X2 = {a,c}
X3 = {a,b,e}
*%

a :- not d.
b :- not a.
b :- e, not d.
c :- a, not e.
c :- b, not e.
d :- not a, not e.
e :- a, d.
e :- b, not c.
```

1.2–b

```
%*
X1 = {a,c,d,f}
X2 = {b,c,d,e}
X3 = {b,e}
*%

a :- not b.
b :- not c.
b :- not f.
c :- a.
c :- d.
d :- c, e.
d :- not b.
e :- not a, not f.
f :- not e.
```

## Exercise 1.3

Reconsider the two logic programs in the previous exercise and find all their answer sets. (HINT: Avoid considering all of the possible interpretations.)

## Exercise 1.4

Determine the answer sets of the following normal logic programs:

1.4–a

```
canary(tweety).
penguin(tux).

% bird(X) :- canary(X).
bird(tweety) :- canary(tweety).
bird(tux)    :- canary(tux).
% bird(X) :- penguin(X).
bird(tweety) :- penguin(tweety).
bird(tux)    :- penguin(tux).

% walks(X) :- penguin(X).
walks(tweety) :- penguin(tweety).
walks(tux)    :- penguin(tux).
% walks(X) :- bird(X), not flies(X).
walks(tweety) :- bird(tweety), not flies(tweety).
walks(tux)    :- bird(tux), not flies(tux).

% flies(X) :- bird(X), not walks(X).
flies(tweety) :- bird(tweety), not walks(tweety).
flies(tux)    :- bird(tux), not walks(tux).
```

1.4–b

```
num(1). num(2). num(3).

% lucky(X) :- num(X), not unlucky(X).
lucky(1) :- num(1), not unlucky(1).
lucky(2) :- num(2), not unlucky(2).
lucky(3) :- num(3), not unlucky(3).

% unlucky(X) :- num(X), not lucky(X).
unlucky(1) :- num(1), not lucky(1).
unlucky(2) :- num(2), not lucky(2).
unlucky(3) :- num(3), not lucky(3).

% p :- lucky(X), lucky(Y), X != Y, not p.
p :- lucky(1), lucky(2), not p.
p :- lucky(1), lucky(3), not p.
p :- lucky(2), lucky(3), not p.
```